

## FCAI ANU Materials – Semi Seniors – Chapter 6 – Visual Programming

### Chapter 6: Arrays and Lists - Multiple Choice Questions

#### Section 1: Array Fundamentals (Questions 1-11)

##### 1. What is an array?

- a) A group of items of different data types stored together.
- b) A group of items of the same data type stored together in memory.
- c) A single variable that can hold multiple names.
- d) A method for processing files.

**Answer: b**

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##### 2. How do you declare an array in C# that can hold integer values?

- a) `int numbersArray;`
- b) `array numbersArray = new int[6];`
- c) `int[] numbersArray;`
- d) `int{} numbersArray;`

**Answer: c**

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##### 3. What does the `new` keyword do when creating an array?

- a) Declares a new variable.
- b) Creates a new object in memory and returns a reference to it.
- c) Initializes all array elements to zero.
- d) Imports a new namespace.

**Answer: b**

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##### 4. Which of the following is a valid array declaration and initialization?

- a) `int[] numbers = new int(6);`
- b) `int numbers = new int[6];`
- c) `int[] numbers = new int[6];`
- d) `int{} numbers = new int[6];`

**Answer: c**

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**5. What is the preferred practice for specifying an array's size?**

- a) Using a literal number like 6.
- b) Using a named constant.
- c) Using a string variable.
- d) Leaving it blank.

**Answer: b**

---

**6. Given the code `decimal[] amounts = new decimal[3];`, what is the index of the last element?**

- a) 1
- b) 2
- c) 3
- d) 4

**Answer: b**

---

**7. How do you assign a value from a `TextBox` named `amount1TextBox` to the first element of a decimal array named `amounts`?**

- a) `amounts[0] = amount1TextBox.Text;`
- b) `amounts[1] = decimal.Parse(amount1TextBox.Text);`
- c) `amounts[0] = decimal.Parse(amount1TextBox.Text);`
- d) `amounts = decimal.Parse(amount1TextBox.Text);`

**Answer: c**

---

**8. What is an initialization list in the context of arrays?**

- a) A list of array names.
- b) A list of methods that can be used on an array.
- c) A series of values inside braces used to initialize an array when it's created.
- d) A list of array indices.

**Answer: c**

---

**9. Which of the following correctly initializes a string array called days?**

- a) string days = {"Sun", "Mon"};
- b) string[] days = new string[] {"Sun", "Mon"};
- c) string[] days = new string2;
- d) string days[] = {"Sun", "Mon"};

**Answer: b**

---

**10. What property of an array returns the number of elements it can hold?**

- a) .Size
- b) .Count
- c) .Capacity
- d) .Length

**Answer: d**

---

**11. What will MessageBox.Show(temperatures.Length); display if temperatures is declared as double[] temperatures = new double[25];?**

- a) 0
- b) 24
- c) 25
- d) 26

**Answer: c**

---

## **Section 2: Working with Arrays (Questions 12-23)**

**12. In the "Numbers application" example, what class is used to generate random numbers?**

- a) Randomizer
- b) Random
- c) Rand

- d) GenerateRandom

**Answer: b**

---

**13. What is the purpose of the for loop in the generateButton\_Click method of the Numbers application?**

- a) To display the numbers in Label controls.
- b) To create the array.
- c) To fill the array with random numbers.
- d) To exit the application.

**Answer: c**

---

**14. How does a foreach loop differ from a for loop when iterating through an array?**

- a) A foreach loop can only be used with string arrays.
- b) A foreach loop does not require a counter variable; it iterates automatically for each element.
- c) A foreach loop is faster than a for loop.
- d) A foreach loop can only be used with two-dimensional arrays.

**Answer: b**

---

**15. What is the correct syntax for a foreach loop that iterates over an integer array named scores?**

- a) foreach (int score in scores)
- b) for (int score : scores)
- c) foreach (scores as int score)
- d) for each (int score in scores)

**Answer: a**

---

**16. In the file writing example, which class is used to write data to a file?**

- a) StreamReader
- b) FileWriter

- c) StreamWriter
- d) TextWriter

**Answer: c**

---

**17. What is the purpose of the File.CreateText method?**

- a) To open an existing file for reading.
- b) To create a new file and get a StreamWriter object.
- c) To delete a file.
- d) To check if a file exists.

**Answer: b**

---

**18. When reading from a file into an array, what condition does the while loop typically check to avoid reading past the end of the file?**

- a) while (index < array.Length)
- b) while (!inputFile.EndOfStream && index < array.Length)
- c) while (inputFile.HasText)
- d) while (index > 0)

**Answer: b**

---

**19. How are arrays passed to methods in C#?**

- a) By value.
- b) By reference.
- c) By copy.
- d) It depends on the data type.

**Answer: b**

---

**20. In Example 3 (Array Argument), what is the output after calling the SetToZero method?**

- a) 1, 2, 3
- b) 0, 0, 0

- c) 1, 0, 0
- d) 0, 2, 3

**Answer: b**

---

**21. Why did the numbers array change in Example 3 after being passed to SetToZero?**

- a) Because the array was returned from the method.
- b) Because arrays are passed by value.
- c) Because arrays are passed by reference, so the method works directly on the original array.
- d) Because of a compiler error.

**Answer: c**

---

**22. In Example 4 (Change Array Demo 1), why did the original numbers array remain unchanged after calling ChangeArray?**

- a) Because the method used the ref keyword.
- b) Because the method created a new array and assigned it to the parameter variable, breaking its reference to the original argument.
- c) Because the array was passed by value.
- d) Because the method only changed the elements, not the array itself.

**Answer: b**

---

**23. What keyword, when used with an array parameter, allows a method to change the reference of the original array variable?**

- a) out
- b) params
- c) ref
- d) in

**Answer: c**

---

## **Section 3: Array Algorithms (Questions 24-33)**

**24. What does the sequential search algorithm do?**

- a) Sorts the array in ascending order.
- b) Uses a loop to step through an array sequentially until it finds the value it is looking for.
- c) Randomly searches the array for a value.
- d) Searches the array starting from the end.

**Answer: b**

---

**25. In the SequentialSearch method, what value is returned if the search value is not found?**

- a) 0
- b) -1
- c) null
- d) false

**Answer: b**

---

**26. In the American Colonies application, what is the purpose of the SequentialSearch method?**

- a) To sort the list of colonies.
- b) To check if a user-selected state is in the array of original colonies.
- c) To display the colonies in a ListBox.
- d) To randomly select a colony.

**Answer: b**

---

**27. Why can't you use the == operator to compare the contents of two arrays for equality?**

- a) Because the == operator is only for strings.
- b) Because it compares the reference variables, not the contents of the arrays.
- c) Because it always returns true.
- d) Because it causes a compiler error.

**Answer: b**

---

**28. To compare the contents of two arrays, you must:**

- a) Use the .Equals() method.
- b) Compare their .Length properties.
- c) Compare each corresponding element in the arrays.
- d) Use the Array.Compare() method.

**Answer: c**

---

**29. In the Test Average application, what does the Highest method do?**

- a) Sorts the array.
- b) Finds the first element in the array.
- c) Finds the largest value in the array.
- d) Calculates the average of the array.

**Answer: c**

---

**30. How is the average calculated in the Average method of the Test Average application?**

- a) By dividing the total by 5.
- b) By dividing the total by the number of elements in the array.
- c) By using the Math.Average function.
- d) By finding the middle value.

**Answer: b**

---

**31. What is the first step in the selection sort algorithm?**

- a) Locate the largest value and move it to the last element.
- b) Locate the smallest value and move it to element 0.
- c) Compare the first two elements.
- d) Reverse the array.

**Answer: b**

---

**32. In the selection sort, after the smallest value is moved to element 0, where does the next scan begin?**

- a) Element 0
- b) Element 1
- c) The last element
- d) A random element

**Answer: b**

---

**33. In the SelectionSort method, what is the purpose of the Swap method?**

- a) To find the smallest value.
- b) To display the array.
- c) To exchange the values of two variables.
- d) To create a new array.

**Answer: c**

---

## **Section 4: Two-Dimensional Arrays (Questions 34-42)**

**34. How many size declarators are required to declare a two-dimensional array?**

- a) One
- b) Two
- c) Three
- d) None

**Answer: b**

---

**35. What is the correct way to declare a two-dimensional array with 3 rows and 4 columns of integers?**

- a) `int[3,4] scores;`
- b) `int[][] scores = new int[3,4];`
- c) `int[,] scores = new int[3,4];`
- d) `int scores[3][4];`

**Answer: c**

---

**36. How do you access the element in the second row and third column of a 2D array named grid?**

- a) grid[1, 2]
- b) grid[2, 3]
- c) grid(1, 2)
- d) grid[2][3]

**Answer: a**

---

**37. What is the common control structure used to process a two-dimensional array?**

- a) if statement
- b) switch statement
- c) Nested loops
- d) do-while loop

**Answer: c**

---

**38. How is a two-dimensional array implicitly initialized?**

- a) With a single set of braces {}.
- b) With nested sets of braces { {}, {} }.
- c) With parentheses ().
- d) It cannot be implicitly initialized.

**Answer: b**

---

**39. In the Seating Chart application, what is the data type of the prices array?**

- a) int
- b) double
- c) decimal
- d) string

**Answer: c**

---

**40. What does the using statement ensure when working with a StreamWriter?**

- a) The file is opened for appending.
- b) The StreamWriter object is properly closed and disposed of, even if an exception occurs.
- c) The file is created if it doesn't exist.
- d) The data is encrypted.

**Answer: b**

---

**41. To append text to an existing file using StreamWriter, what must you do?**

- a) Use the Append method.
- b) Pass true as the second argument to the StreamWriter constructor.
- c) Use File.AppendText instead.
- d) Both b and c are correct.

**Answer: d**

---

**42. In Example 10, what does the btnShowArray\_Click method do?**

- a) Saves the array to a file.
- b) Loads the array from a file.
- c) Displays the contents of the 2D array and the average for each row in a ListBox.
- d) Clears the ListBox.

**Answer: c**

---

## **Section 5: Jagged Arrays (Questions 43-47)**

**43. What is the key difference between a two-dimensional array and a jagged array?**

- a) A two-dimensional array can only hold integers.
- b) A jagged array is an array of arrays, where each inner array can have a different length.
- c) A jagged array must be rectangular.
- d) A two-dimensional array uses parentheses ().

**Answer: b**

---

**44. How do you declare a jagged array of integers?**

- a) `int[,] jaggedArray;`
- b) `int[][] jaggedArray;`
- c) `int{} jaggedArray;`
- d) `int jaggedArray[][];`

**Answer: b**

---

**45. Given `int[][] jagged = new int[2][];`, how do you initialize the first inner array to hold 3 integers?**

- a) `jagged[0] = new int[3];`
- b) `jagged[0] = {1, 2, 3};`
- c) `jagged[1] = new int[3];`
- d) `jagged = new int[2][3];`

**Answer: a**

---

**46. How do you access the third element of the second array in a jagged array named `data`?**

- a) `data[2][3]`
- b) `data[1][2]`
- c) `data[3][2]`
- d) `data[2][1]`

**Answer: b**

---

**47. What does `jaggedArray.Length` return for a jagged array?**

- a) The total number of elements in all inner arrays.
- b) The length of the first inner array.
- c) The number of rows (the number of inner arrays).
- d) An error.

**Answer: c**

---

## **Section 6: Lists (Questions 48-59)**

**48. What is a major advantage of a List over an array?**

- a) Lists are stored in a different part of memory.
- b) Lists do not require a using directive.
- c) Lists can automatically resize as items are added or removed.
- d) Lists can only hold strings.

**Answer: c**

---

**49. How do you create a List that can hold strings?**

- a) `List stringList = new List();`
- b) `List<string> stringList = new List<string>();`
- c) `string[] stringList = new List<>();`
- d) `ArrayList stringList = new ArrayList();`

**Answer: b**

---

**50. Which method is used to add an item to a List?**

- a) `.Add()`
- b) `.Insert()`
- c) `.Append()`
- d) `.Push()`

**Answer: a**

---

**51. What property of a List tells you how many items it contains?**

- a) `.Size`
- b) `.Length`
- c) `.Capacity`
- d) `.Count`

**Answer: d**

---

**52. How can you access the second element in a `List<string>` named `nameList`?**

- a) nameList(1)
- b) nameList[1]
- c) nameList.Get(1)
- d) nameList.Item(1)

**Answer: b**

---

**53. How do you remove the item at index 2 from a List?**

- a) list.Remove(2);
- b) list.Delete(2);
- c) list.RemoveAt(2);
- d) list.Clear(2);

**Answer: c**

---

**54. What is the difference between the Remove and RemoveAt methods?**

- a) Remove takes an index, RemoveAt takes an object.
- b) Remove takes an object, RemoveAt takes an index.
- c) They are identical.
- d) Remove is faster than RemoveAt.

**Answer: b**

---

**55. What does the Remove method return?**

- a) The item that was removed.
- b) The index of the item that was removed.
- c) A boolean indicating whether the item was found and removed.
- d) Nothing (void).

**Answer: c**

---

**56. What does the IndexOf method return if the item is not found in the List?**

- a) 0
- b) -1

- c) null
- d) false

**Answer: b**

---

**57. In the statement `int pos = nameList.IndexOf("Diane", 2);`, what does the 2 represent?**

- a) The number of times to search.
- b) The starting index for the search.
- c) The length of the string to search for.
- d) The ending index for the search.

**Answer: b**

---

**58. In the Test Score List application (Example 12), what is the purpose of the `ReadScores` method?**

- a) To calculate the average score.
- b) To display scores in the `ListBox`.
- c) To read scores from a file into a `List`.
- d) To count the above-average scores.

**Answer: c**

---

**59. How does the `AboveAverage` method determine the number of above-average scores?**

- a) It compares each score to a fixed value of 80.
- b) It first calls the `Average` method to get the average, then counts scores greater than that average.
- c) It sorts the list and counts the top half.
- d) It uses the `Max` method.

**Answer: b**

---

## **Section 7: Quiz Application (Questions 60-65)**

**60. In the Quiz application (Example 13), what data structure is used to store the questions?**

- a) A two-dimensional array
- b) A jagged array
- c) A single-dimensional string array
- d) A List of strings

**Answer: c**

---

**61. How are the answer options for the quiz stored?**

- a) In a single-dimensional array.
- b) In a List of strings.
- c) In a two-dimensional string array.
- d) In a jagged integer array.

**Answer: c**

---

**62. What does the correctAnswers array hold?**

- a) The text of the correct answers.
- b) The indices of the correct options for each question.
- c) The scores for each question.
- d) The questions themselves.

**Answer: b**

---

**63. What is the role of the LoadQuestion method?**

- a) To check the user's answer.
- b) To display the current question and its options in the form controls.
- c) To move to the next question.
- d) To calculate the final score.

**Answer: b**

---

**64. In the btnSubmit\_Click method, what happens if the user's selected answer matches the value in the correctAnswers array for the current question?**

- a) The text "Incorrect!" is displayed in red.

- b) The application closes.
- c) The text "Correct!" is displayed in green.
- d) The next question loads automatically.

**Answer: c**

---

**65. What happens when the user clicks the "Next" button after the last question?**

- a) The application crashes.
- b) The first question loads again.
- c) A completion message is displayed, and the input controls are hidden/disabled.
- d) Nothing happens.

**Answer: c**

---

## **Section 8: Additional Concepts (Questions 66-100)**

**66. What is the default value for elements in a numeric array (e.g., int[])?**

- a) null
- b) -1
- c) 0
- d) 1

**Answer: c**

---

**67. What happens if you try to access an array element with an index that is negative or greater than or equal to the array's length?**

- a) The value 0 is returned.
- b) The last element is accessed.
- c) An `IndexOutOfRangeException` occurs.
- d) The program compiles but runs slowly.

**Answer: c**

---

**68. The expression `int[] numbers = { 10, 20, 30 };` is an example of:**

- a) Array declaration
- b) Array declaration and implicit initialization
- c) Array resizing
- d) A compiler error

**Answer: b**

---

**69. Which loop is best suited when you need the index of each element in an array?**

- a) foreach loop
- b) while loop
- c) do-while loop
- d) for loop

**Answer: d**

---

**70. What is the result of the following code? `int[] arr = new int[1];`  
`MessageBox.Show(arr[0].ToString());`**

- a) "1"
- b) "0"
- c) "null"
- d) An error

**Answer: b**

---

**71. When you pass an array to a method and the method modifies an element, what happens to the original array?**

- a) The original array is unchanged.
- b) The original array is changed because the method receives a reference to the array.
- c) A copy of the array is modified.
- d) It causes a compilation error.

**Answer: b**

---

**72. What is the purpose of the Swap method in the selection sort algorithm?**

- a) To find the minimum value.
- b) To iterate through the array.
- c) To exchange the positions of two elements.
- d) To display the sorted array.

**Answer: c**

---

**73. In a two-dimensional array, the first dimension typically represents:**

- a) Columns
- b) Rows
- c) Depth
- d) Pages

**Answer: b**

---

**74. What does `myArray.GetLength(0)` return for a 2D array?**

- a) The number of columns.
- b) The total number of elements.
- c) The number of rows.
- d) The length of the first element.

**Answer: c**

---

**75. To process every element in a 2D array, you commonly use:**

- a) A single for loop.
- b) A foreach loop and a while loop.
- c) Two nested for loops.
- d) A switch statement.

**Answer: c**

---

**76. What is a key characteristic of a jagged array?**

- a) It must be rectangular.
- b) All its rows must have the same length.

- c) Its rows can be of different lengths.
- d) It can only have two dimensions.

**Answer: c**

---

**77. How do you initialize a specific inner array of a jagged array?**

- a) `jaggedArray[0] = new int[] {1, 2, 3};`
- b) `jaggedArray = {1, 2, 3};`
- c) `jaggedArray[0][0] = new int[3];`
- d) `jaggedArray(0) = [1,2,3];`

**Answer: a**

---

**78. What is the correct way to access an element in a jagged array?**

- a) `jaggedArray[0, 1]`
- b) `jaggedArray[0][1]`
- c) `jaggedArray(0)(1)`
- d) `jaggedArray.GetValue(0, 1)`

**Answer: b**

---

**79. Which of the following is a disadvantage of arrays compared to Lists?**

- a) Arrays have a fixed size.
- b) Arrays can hold different data types.
- c) Arrays are faster.
- d) Arrays use more memory.

**Answer: a**

---

**80. What is the correct way to initialize a `List<int>` with some initial values?**

- a) `List<int> list = new List<int>(1, 2, 3);`
- b) `List<int> list = new List<int>() {1, 2, 3};`
- c) `List<int> list = {1, 2, 3};`
- d) `List<int> list = new List<int>().Add(1).Add(2).Add(3);`

**Answer: b**

---

**81. What happens if you call RemoveAt with an index that is out of range?**

- a) The last item is removed.
- b) The first item is removed.
- c) An ArgumentOutOfRangeException occurs.
- d) Nothing happens.

**Answer: c**

---

**82. The IndexOf("A", 2, 5) method searches for "A":**

- a) From the start of the list to the end.
- b) From index 2 to index 5 (inclusive).
- c) From index 2 to index 5 (exclusive).
- d) Only at index 2 and index 5.

**Answer: b**

---

**83. In the context of the List collection, what does "pass by reference" mean for the methods that modify the list?**

- a) You must use the ref keyword.
- b) The methods work on a copy of the list.
- c) The methods directly modify the original list object.
- d) The list is passed as a value.

**Answer: c**

---

**84. In the Quiz application, what is the purpose of the currentQuestionIndex variable?**

- a) To store the user's score.
- b) To keep track of which question is currently being displayed.
- c) To hold the correct answers.
- d) To count the number of questions.

**Answer: b**

---

**85. What does the ClearSelection method in the Quiz application do?**

- a) Resets the currentQuestionIndex to 0.
- b) Unchecks all the RadioButton controls.
- c) Clears the lblQuestion text.
- d) Closes the form.

**Answer: b**

---

**86. If File.OpenText cannot find the specified file, what will happen?**

- a) It will create a new file.
- b) An exception will be thrown.
- c) It will return null.
- d) The program will ignore it.

**Answer: b**

---

**87. What is the benefit of using a List for the test scores in Example 12 instead of an array?**

- a) The List automatically calculates the average.
- b) The List can dynamically grow to hold all the scores from the file, regardless of how many there are.
- c) The List is always faster.
- d) The List can only hold unique scores.

**Answer: b**

---

**88. What is the scope of the scoresList variable in the getScoresButton\_Click method?**

- a) It is a global variable.
- b) It is local to the method.
- c) It is passed from the ReadScores method.
- d) It is static.

**Answer: b**

---

**89. What is the purpose of the try-catch block in file handling code?**

- a) To make the code run faster.
- b) To handle potential exceptions (errors) that may occur during file operations.
- c) To automatically close the file.
- d) To validate user input.

**Answer: b**

---

**90. In the selection sort, how many passes are needed for an array of n elements?**

- a) 1
- b) n
- c) n-1
- d) n/2

**Answer: c**

---

**91. What is the time complexity of the sequential search algorithm in the worst case?**

- a)  $O(1)$
- b)  $O(\log n)$
- c)  $O(n)$
- d)  $O(n^2)$

**Answer: c**

---

**92. What is the time complexity of the selection sort algorithm?**

- a)  $O(n)$
- b)  $O(n \log n)$
- c)  $O(n^2)$
- d)  $O(1)$

**Answer: c**

---

**93. What does the `!inputFile.EndOfStream` condition mean?**

- a) The file is empty.
- b) The file is at the beginning.
- c) The file pointer has not reached the end of the file.
- d) The file is write-protected.

**Answer: c**

---

**94. In the `getNamesButton_Click` method of Example 1, how are the names displayed?**

- a) In a single `MessageBox` showing all names.
- b) In a `ListBox`.
- c) In three separate `MessageBoxes`.
- d) In a `Label` control.

**Answer: c**

---

**95. What is the initial value of the position variable in the `SequentialSearch` method?**

- a) 0
- b) 1
- c) -1
- d) null

**Answer: c**

---

**96. What is the purpose of the found flag in the `SequentialSearch` method?**

- a) To count the number of comparisons.
- b) To indicate when the search value has been found so the loop can stop.
- c) To store the found value.
- d) To handle errors.

**Answer: b**

---

**97. In the `ChangeArray` method of Example 5 (with `ref`), what does `iArray = new int[NEW_SIZE];` do?**

- a) It changes the local parameter `iArray` to reference a new array.

- b) It changes the original numbers array in the calling method to reference a new array.
- c) It creates a new array but doesn't assign it.
- d) It causes an error.

**Answer: b**

---

**98. What is the output of the array comparison code if firstArray is {2,4,6,8,10} and secondArray is {2,4,6,8,10}?**

- a) "The arrays are equal."
- b) "The arrays are not equal."
- c) An error occurs.
- d) The output is undefined.

**Answer: a**

---

**99. In the Average method for a List<int>, why is (double) used in the calculation?**

- a) To convert the total to a double and ensure floating-point division.
- b) To round the result.
- c) To avoid an exception.
- d) To format the output.

**Answer: a**

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**100. What is the final state of the array after the selection sort process is complete?**

- a) The array is reversed.
- b) The array is sorted in ascending order.
- c) The array is sorted in descending order.
- d) The array is shuffled randomly.

**Answer: b**

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**Wish you all the best!**  
**Fathy.**